

Cluster 10269

To view the latest Map of Science clusters and data, visit the live Map at sciencemap.eto.tech.

Subjects

Research fields: Mathematics, Physics, Computer science

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Combinatorics, Discrete mathematics, Pure mathematics

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: finite set, distance problem, product sets, plane point set, Furstenberg sets

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

0.55% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to cybersecurity

Vital signs

373 recent articles in the cluster

373 articles in this cluster were published in the five year period ending in 2023.

Average article is **16.19** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **20.16th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 20.16% of other clusters.

Citation rating: **39.32th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 39.32% of all other articles published in the same year.

93.30% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 224 of 373 articles. 180 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
United States	62	0.95
Russia	27	0.43
United Kingdom	25	1.20
Vietnam	25	0.82
Austria	21	0.93
Canada	20	0.46
Australia	17	0.51
Hungary	15	0.62
Israel	13	0.45
South Korea	11	0.65

Top funding countries

Data available for 137 of 373 articles. 119 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
United States	35	1.20
Russia	19	0.60
Canada	18	0.68
Austria	16	1.01
China	12	0.69
Australia	12	0.64
Switzerland	9	1.49
South Korea	9	0.77
Vietnam	8	0.82
Israel	6	0.33

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

[On Sum Sets and Convex Functions](#)

2022: The Electronic Journal of Combinatorics. 6 citations.

[On the pinned distances problem in positive characteristic](#)

2020: arXiv: Combinatorics. 31 citations.

Some new results on the higher energies I

2023: arXiv (Cornell University). 38 citations.

On the Pinned Distances Problem over Finite Fields

2020: arXiv (Cornell University). 28 citations.

Energy estimates in sum-product and convexity problems

2021: arXiv: Combinatorics. 9 citations.

Attaining the exponent $5/4$ for the sum-product problem in finite fields

2021: arXiv: Combinatorics. 10 citations.

[The quotient set of the quadratic distance set over finite fields](#)

2023: arXiv (Cornell University). 6 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

[The Finite Field Distance Problem](#)

2021: The Carus mathematical monographs. 2 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[On the pinned distances problem in positive characteristic](#)

2020: arXiv: Combinatorics. 31 citations.

Some new results on the higher energies I

2023: arXiv (Cornell University). 38 citations.

On the Pinned Distances Problem over Finite Fields

2020: arXiv (Cornell University). 28 citations.

Furstenberg sets estimate in the plane

2023: arXiv (Cornell University). 23 citations.

[Query complexity and the polynomial Freiman-Ruzsa conjecture](#)

2020: arXiv: Number Theory. 18 citations.

Attaining the exponent $5/4$ for the sum-product problem in finite fields

2021: arXiv: Combinatorics. 10 citations.

Projections, Furstenberg sets, and the ABC sum-product problem

2023: arXiv (Cornell University). 20 citations.

[Polynomial Methods and Incidence Theory](#)

2022. 23 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 366 out of 373 articles. 258 of these articles came from these sources.

Source	Articles (last five years)
arXiv (Cornell University)	173
arXiv: Combinatorics	37
arXiv: Number Theory	12
Математические заметки	9
Finite Fields and Their Applications	6
arXiv: Classical Analysis and ODEs	5
arXiv	5
Acta Arithmetica	4
arXiv.org	4
Combinatorica	3

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 369 of 373 articles. 101 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Thang Pham	34	1.88
Alex Iosevich	20	1.15
Igor E. Shparlinski	18	1.18
Doowon Koh	15	1.44
Ilya D. Shkredov	13	4.72
Oliver Roche-Newton	13	1.03
Thang V. Pham	13	0.76
Steven Senger	12	1.54
Oliver Roche-Newton	11	1.07
Brian McDonald	10	1.37

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 228 of 373 articles. 94 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
University of Rochester - United States	19	0.87
Vietnam National University, Hanoi - Vietnam	18	1.03
UNSW Sydney - Australia	14	0.55
University of British Columbia - Canada	12	0.53
Johann Radon Institute for Computational and Applied Mathematics - Austria	12	1.21
Steklov Mathematical Institute - Russia	11	0.83
Chungbuk National University - South Korea	9	0.8
Russian Academy of Sciences - Russia	8	0.35
Alfréd Rényi Institute of Mathematics - Hungary	8	0.73
Eötvös Loránd University - Hungary	8	0.49

Organization types

This table covers the 221 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 373 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
0.00%	44.67%	7.99%	2.66%

Top industry organizations

No data available.

Top funders

Data available for 156 out of 373 recent articles. 105 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Science Foundation - United States	31	1.25
Natural Sciences and Engineering Research Council - Canada	17	0.49
FWF Austrian Science Fund - Austria	16	1.01
Australian Research Council - Australia	12	0.64
Russian Science Foundation - Russia	10	0.38
National Natural Science Foundation of China	9	0.64
Swiss National Science Foundation - Switzerland	9	1.49
National Research, Development & Innovation Office (NRDIO) - Hungary	8	0.49
European Research Council (ERC)	7	0.29
Planning and Budgeting Committee of the Council of Higher Education - Israel	6	0.33

Funder types

This table covers the 136 articles in this cluster from the last five years with data about their funding organization types (out of 373 articles total).

Commercial	Education	Nonprofit	Government
0.00%	12.44%	19.62%	67.94%

Patents and industry

- 1 patents cite articles in this cluster (14.76th percentile among all clusters)
- 0.04% of articles in the cluster are cited by patents
- 0% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
ERROR-DETECTION SCHEMES FOR ANALOG CONTENT-ADDRESSABLE MEMORIES	1

Top patent assignees

Patent assignee	Same-cluster citations
HEURON CO., LTD.	1

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
3540	16	Hausdorff dimension, multifractal Hausdorff, intermediate dimensions, Borel probability measure, general Hausdorff measure
48572	14	Compact Lie Groups, Cayley graphs, Lie group, finite simple groups, approximate lattices
10997	10	inverse theorem, polynomial, abelian group, ergodic averages, subset
2564	6	Voronoi diagram, query time, data structure, Shortest Path Problem, lower bound
9562	4	Takeya sets, Fourier Restriction, maximal function, Fourier integral operators, Hausdorff dimension
53268	3	rational Diophantine triples, Diophantine quadruples, simultaneous Pell equations, positive integers, generalized Diophantine
21959	3	crossing number, Graph Drawing, edge crossings, Planar Graphs, vertices
24429	3	Turán number, generalized Turán number, maximum number, Turán graph, n-vertex F-free graph
8470	2	Fourier coefficients, cusp form, holomorphic cusp form, Hecke eigenvalues, Hecke Maass form
29896	2	Lower Bounds, Polynomial Identity Testing, polynomial time algorithms, deterministic polynomial time, Algebraic Circuits

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
3540	29	Hausdorff dimension, multifractal Hausdorff, intermediate dimensions, Borel probability measure, general Hausdorff measure
10997	25	inverse theorem, polynomial, abelian group, ergodic averages, subset
9562	16	Keakeya sets, Fourier Restriction, maximal function, Fourier integral operators, Hausdorff dimension
48572	11	Compact Lie Groups, Cayley graphs, Lie group, finite simple groups, approximate lattices
8470	8	Fourier coefficients, cusp form, holomorphic cusp form, Hecke eigenvalues, Hecke Maass form
8855	7	random graph, Turán density, Random Turán number, Hypergraphs, Ramsey number problem
2564	6	Voronoi diagram, query time, data structure, Shortest Path Problem, lower bound
41404	5	maximum order complexity, Nth maximum order, automatic sequences, Pseudorandom Binary Sequences, Correlation measure
36466	5	Gauss sums, quadratic Gauss sums, two-term exponential sums, character sums, generalized quadratic Gauss
21959	5	crossing number, Graph Drawing, edge crossings, Planar Graphs, vertices

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
the-set-of-sets/nin	1	2
the-set-of-sets/l1_linfty	1	0
hvds/seq	1	12
anuragsahay/vcdimsquares	1	0
andrewlott99/SP-k-	1	0